1)Digital input signals A, B, C with A as the MSB and C as the LSB are used to realize the Boolean function $F = m_0 + m_2 + m_3 + m_5 + m_7$ where m_i denotes the ith minterm. In addition, F has a don't care for m_1. The simplified expression for F is given by

2) Simplified form of the Boolean function: $F(P, Q, R, S) = P' Q' + P' QS + PQ' R' S' + PQ'RS'$.

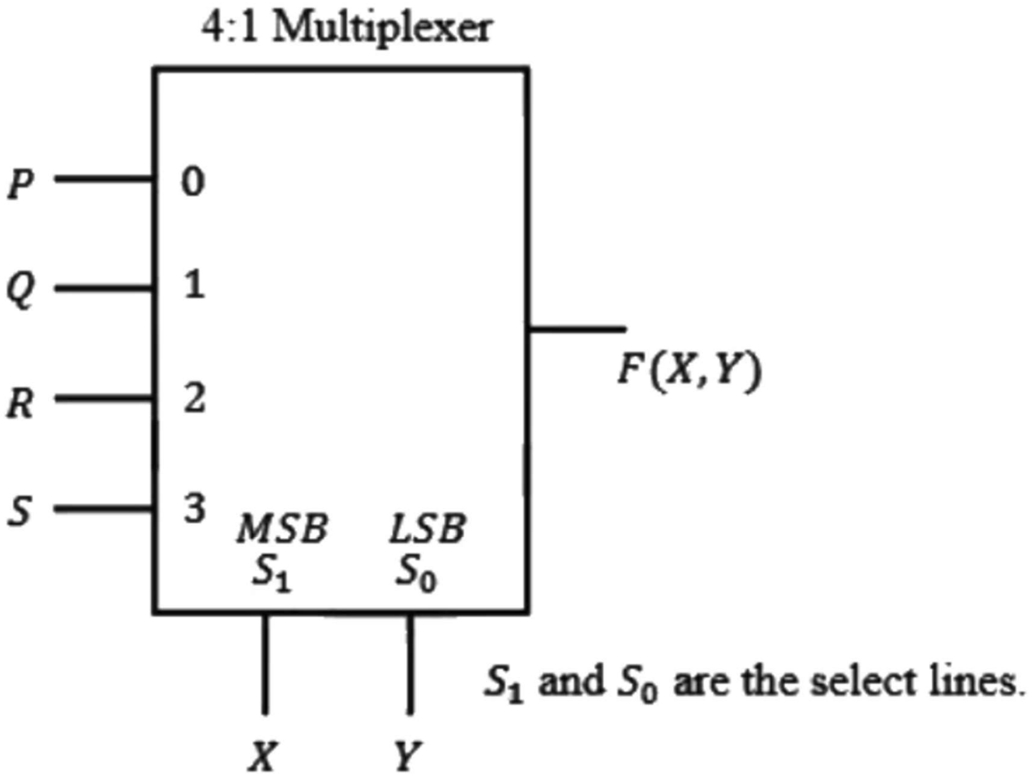
3) F(A, B, C, D) = Π M(0, 1, 3, 4, 5, 7, 9, 11, 12, 13, 14, 15) is a maxterm representation of a Boolean function f(A, B, C, D) where A is the MSB and D is the LSB. The equivalent minimized representation of this function is

4)In the sum of products function f(X, Y, Z) = (2, 3, 4, 5), the prime implicants are

5)The following Karnaugh map represents a function F.   A minimized form of the function F is

X \ F	00	01	11	10
0	1	1	1	0
1	0	0	1	0

6)To obtain the Boolean function F(X,Y) = XY + X, the inputs PQRS in the figure should be

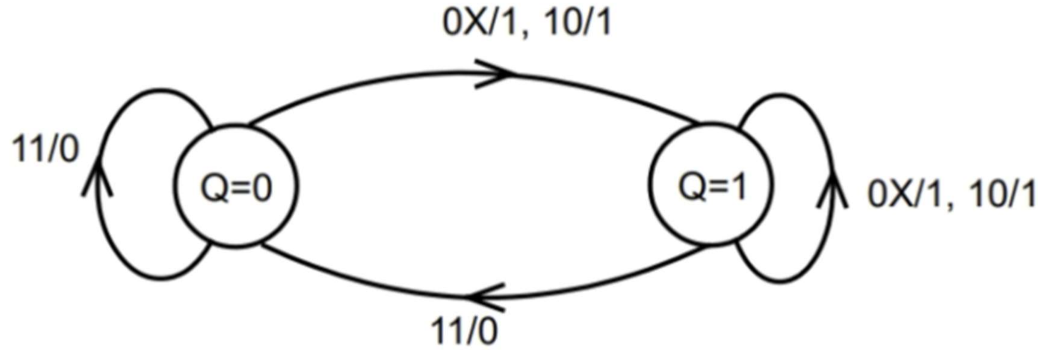


7)A counter is constructed with three D flip-flops. The input-output pairs are named (D0,Q0) (D1 , Q1) and (D2, Q2) where the subscript 0 denotes the least significant bit. The output sequence is desired to be the Gray-code sequence 000, 001, 011, 010, 110, 111, 101 and 100, repeating periodically. Note that the bits are listed in the Q2Q1Q0 format. The combinational logic expression for D1 is

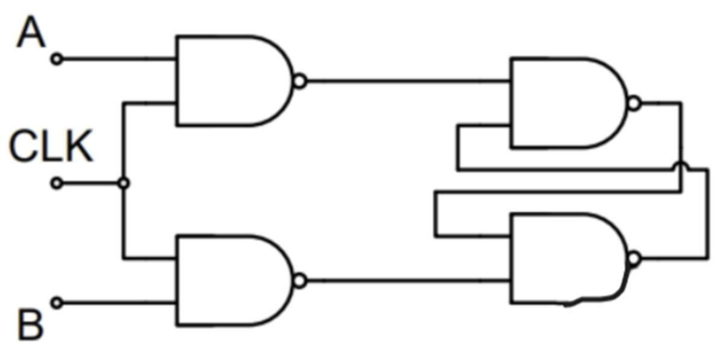
8)The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The number of combinations for which the output is logic 1, is

9) The maximum clock frequency in MHz of a 4-stage ripple counter, utilizing flip-flops, with (round off to one decimal each flip-flop having a propagation delay of 20 ns, is place).

10) A state diagram of a logic gate which exhibits a delay in the output is shown in the figure, where X is the don't care condition, and Q is the output representing the state. The logic gate represented by the state diagram is

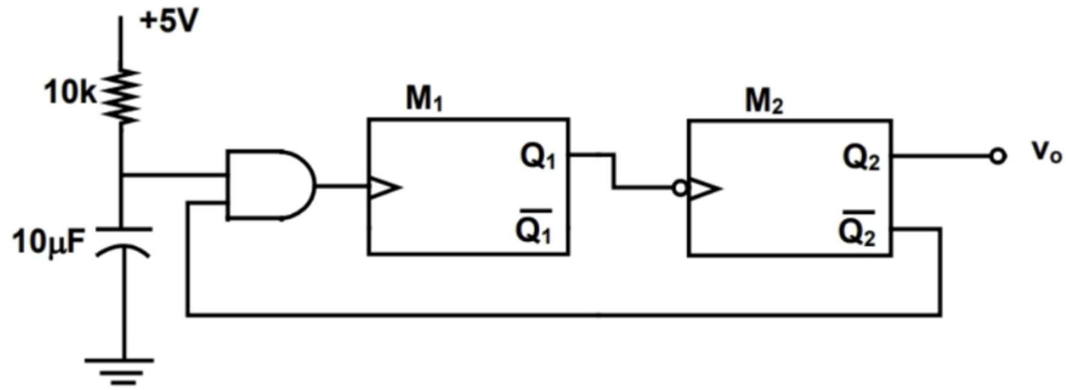


11)Consider the given circuit. In this circuit, the race around occurs when



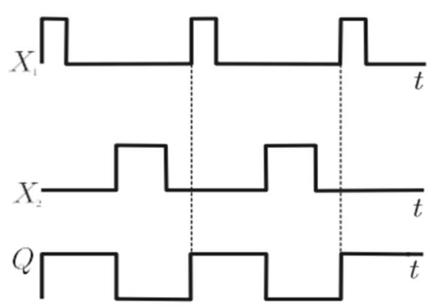
12)

Two monoshot multivibrators, one positive edge triggered (M_1) and another negative edge triggered (M_2) are connected as shown in figure.

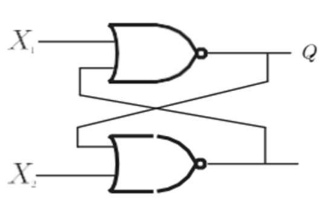


The monoshots M_1 and M_2 when triggered produce pulses of width T_1 and T_2 respectively, where $T_1 > T_2$. The steady state output voltage V_0 of the circuit is

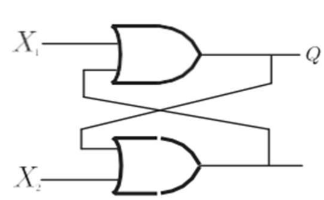
13) Select the circuit which will produce the given output Q for the input signals X1 and X2 given in the figure



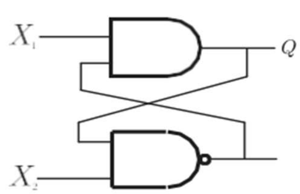
(A)



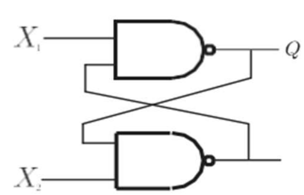
(B)



(C)

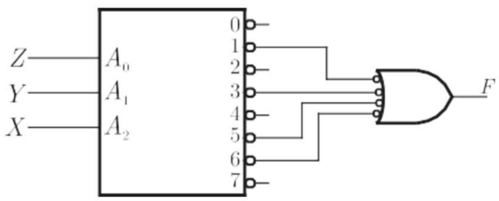


(D)



14)

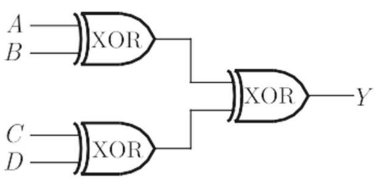
A 3-line to 8-line decoder, with active low outputs, is used to implement a 3-variable Boolean function as shown in the figure



The simplified form of Boolean function $F(A, B, C)$ implemented in 'Product of Sum' form will be

~~15)~~

A, B, C and D are input, and Y is the output bit in the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?



(A) S is always with zero or odd

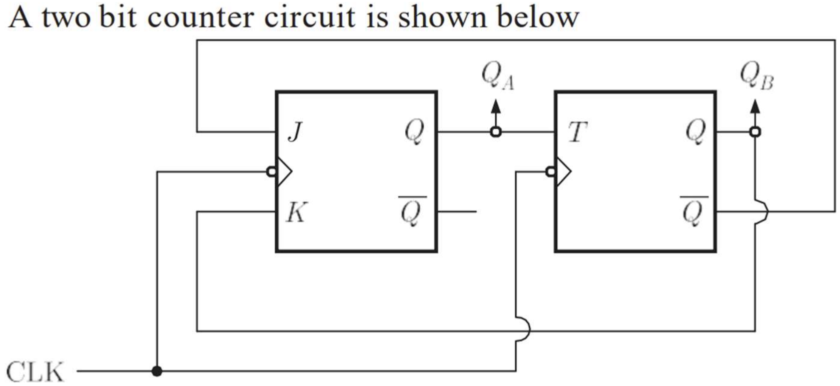
(B) S is always either zero or even

(C) $S = 1$ only if the sum of A, B, C and D is even

(D) $S = 1$ only if the sum of A, B, C and D is odd

~~16)~~

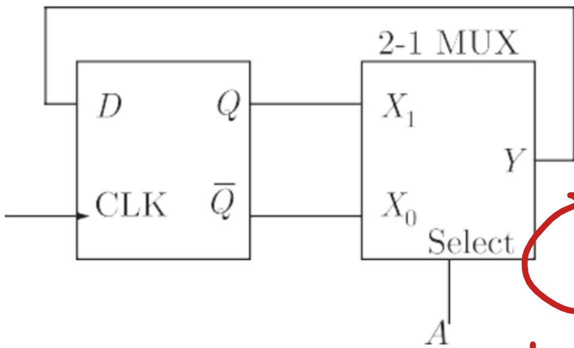
A two bit counter circuit is shown below

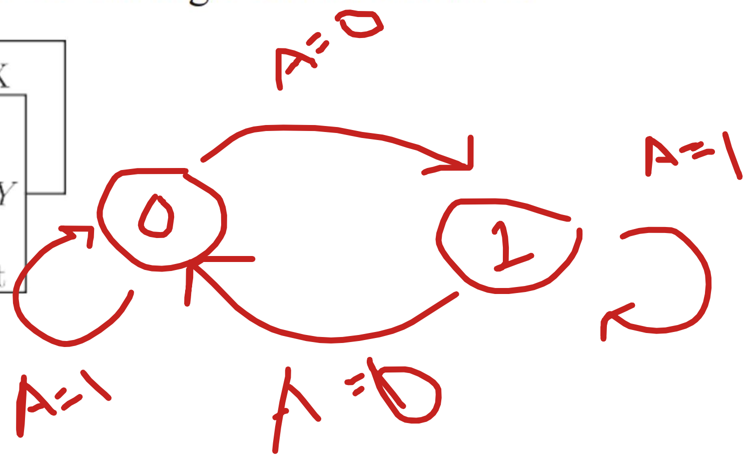


It the state $Q_A\,Q_B$ of the counter at the clock time t_n is '10' then the state $Q_A\,Q_B$ of the counter at $t_n + 3$ (after three clock cycles) will be

17)

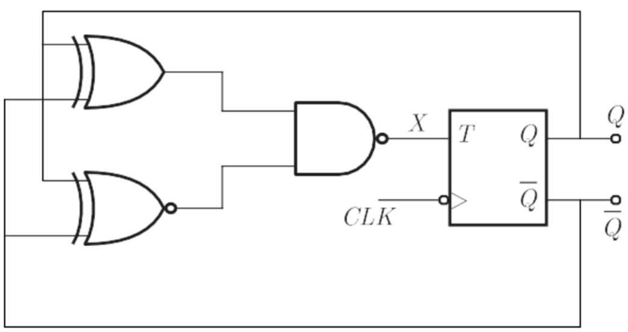
The state transition diagram for the logic circuit shown is





18)

The clock frequency applied to the digital circuit shown in the figure below is 1 kHz. If the initial state of the output of the flip-flop is 0, then the frequency of the output waveform Q in kHz is



(A) 0.25

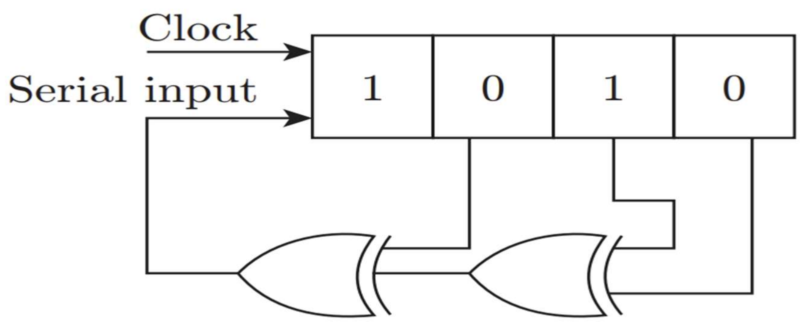
(B) 0.5

(C) 1

(D) 2

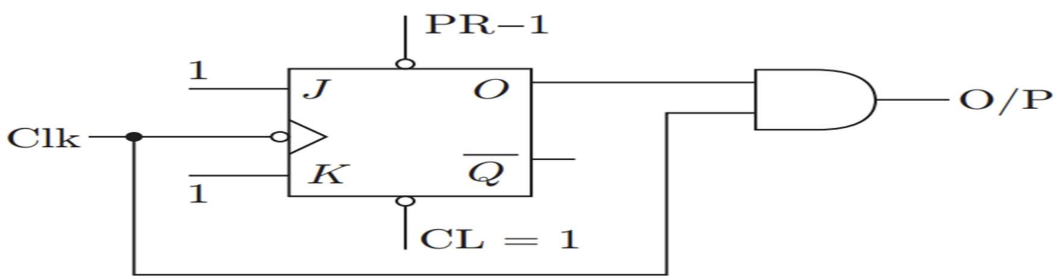
19)

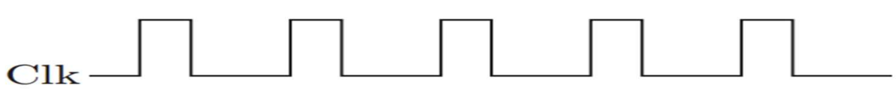
The shift register shown in the given figure is initially loaded with the bit pattern 1010. Subsequently, the shift register is clocked, and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). After how may clock pulses will the content of the shift register become 1010 again?



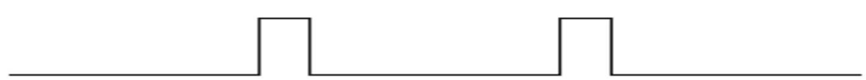
20)

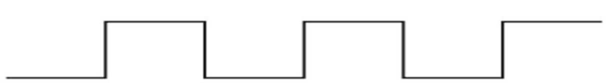
The digital circuit shown in the given figure generates a modified clock pulse at the output. Choose the correct output waveform from the options given below.





(a) 

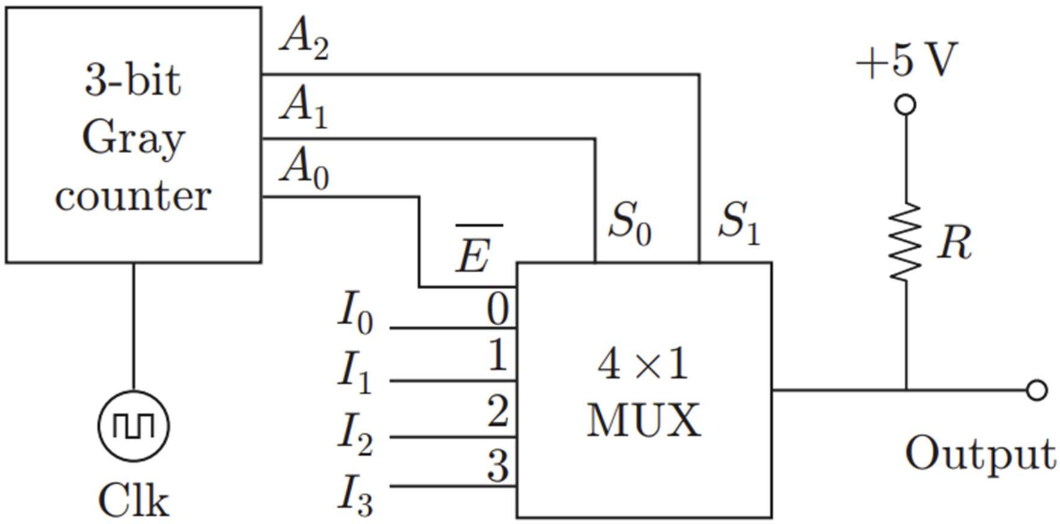
(b) 

(c) 

(d) 

21)

A 3-bit Gray counter is used to control the output of the multiplexer as shown in the figure. The initial state of the counter is $(000)_2$. The output is pulled high. The output of the circuit follows the



Write for the next 5 states at output.

22)A 16 -bit synchronous binary up-counter is clocked with a frequency . The two most significant bits are ORed together to form an output . Measurements shows that is periodic and the duration for which the remains high in each period is 24 ms . The clock frequency clk is MHz.

23)The maximum clock frequency in MHz of a 4-stage ripple counter, utilizing flip-flops, with each flip-flop having a propagation delay of 20 ns, is